

P2505R0 Monadic Functions for

`std::expected`

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Audience: LEWG, LWG

Project: ISO/IEC JTC1/SC22/WG21 14882: Programming Language — C++

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Introduction

With the final plenary vote of [P0798 Monadic Functions for `std::optional`](#) complete, we now have a design inconsistency with `std::expected`. [P0323 `std::expected`](#) is currently in final LWG wording review for C++23. This proposal corrects the inconsistency by adding 3 functions to `std::expected` and is targeted at C++23.

The intent from P0798 clearly states that `std::expected` should gain the same functions:

There is a proposal for `std::expected` which would benefit from many of these same ideas. If the idea to add monadic interfaces to standard library classes on a case-by-case basis is chosen rather than a unified non-member function interface, then compatibility between this proposal and the `std::expected` one should be maintained

Motivation and Scope

The following 3 functions are added to `std::optional`, but are currently not part of `std::expected`.

- `and_then` compose a chain of functions returning an `expected`
- `or_else` returns if it has a value, otherwise it calls a function with the error type
- `transform` apply a function to change the value (and possibly the type)

For example, given the following:

```
using time_expected = expected<boost::posix_time::ptime, std::string>;

time_expected from_iso_str( std::string time )
{
    try {
        ptime t = boost::posix_time::from_iso_string( time );
        return t;
    }
    catch( std::exception& e )
    {
        return unexpected( e.what() + " str was: "s + time);
    }
}

// for use with transform
ptime next_day( boost::posix_time::ptime t )
{
    return t + boost::gregorian::days(1);
}

// for use with or_else
void print_error( std::string error )
{
    cout << error << endl;
}

//valid iso string
const std::string ts ( "20210726T000000" );
```

Before the change we'd write this:

Before

```
time_expected d = from_iso_str( ts );
if (d)
{
    ptime t = next_day( *d );
}
else
{
    print_error( d.error() );
}
```

And after, this:

After

```
// chain a series of functions until there's an error
auto d = from_iso_str ( ts )
        .or_else( print_error )
        .transform( next_day );
```

For more examples see: <https://godbolt.org/z/va5fzx11f>

Design Considerations

Compared with `optional`, `expected` provides the capability for `void` as a function return type. This complicates the design compared to `optional`. As currently specified the `void` specialization of `expected`, `expected<void, T>` removes the `value_or` function since `void` cannot be initialized to anything else. The situation is similar with `transform` when the expected value type is `void`. Thus `transform` should not compile with `expected<void, err>`. The following example illustrates the proposed behavior.

```

using void_expected = expected<void, std::string>;

// for use with or_else
void print_error( std::string error )
{
    cout << error << endl;
}

void_expected doit(int x)
{
    if (x < 5) return {}; //void return
    return unexpected(std::format("X must be less than 5 passed {}", x))
}

// chained in and_then
void_expected doit2()
{
    return {};
}

auto res = doit(1).or_else( print_error ); //res.has_value() == true
res = doit(5).or_else( print_error );      //call print_error
res = doit(1).and_then( doit2 );           //call doit2
//res = doit(5).transform(doit(5));        //compile error
//res = doit(5).value_or({});              //compile error

```

Other Languages

Rust provides a result type with similar functions monadic functions.

<https://doc.rust-lang.org/std/result/enum.Result.html>

Implementations

This has been implemented by Sy Brand

<https://github.com/TartanLlama/expected> with [documentation here](#). The author is unaware of another expected implementation that adds these functions.

Wording

Feature Test Macro

Update the `__cpp_lib_expected` feature test macro from P0323. Although this could be considered optional since `expected` is not yet a c++ feature in any implementations.

Class template `expected` [`expected.expected`]

After `value_or` functions add the new functions:

```
template<class U> constexpr T value_or(U&&) &&;

// [expected.monadic], monadic operations
template <class F> constexpr auto and_then(F&& f) &;
template <class F> constexpr auto and_then(F&& f) &&;
template <class F> constexpr auto and_then(F&& f) const&;
template <class F> constexpr auto and_then(F&& f) const&&;
template <class F> constexpr auto transform(F&& f) &;
template <class F> constexpr auto transform(F&& f) &&;
template <class F> constexpr auto transform(F&& f) const&;
template <class F> constexpr auto transform(F&& f) const&&;
template <class F> constexpr expected or_else(F&& f) &&;
template <class F> constexpr expected or_else(F&& f) const&;
// [expected.mod], modifiers
```

Add a new sub-clause Monadic operations for [`expected.monadic`] between [`expected.observe`] and [`expected.equality`]

1 `template <class F> constexpr auto and_then(F&& f) &;`

2 `template <class F> constexpr auto and_then(F&& f) const&;`

Let U be `invoke_result_t<F, decltype(value())>`.

Constraints: F models `invocable<decltype(value())>`. E and U model `copy_constructible`.

Mandates: `remove_cvref_t<U>` is a specialization of `expected`.

Effects: Equivalent to:

```
if (*this) {
    return invoke(std::forward<F>(f), value());
}
else {
    return *this;
}
```

1 `template <class F> constexpr auto and_then(F&& f) &&;`

2 `template <class F> constexpr auto and_then(F&& f) const&&;`

Let `U` be `invoke_result_t<F, decltype(std::move(value()))>`.

Constraints: `F` models `invocable<decltype(std::move(value()))>`. `E` and `U` model `move_constructible`.

Mandates: `remove_cvref_t<U>` is a specialization of `expected`.

Effects: Equivalent to:

```
if (*this) {
    return invoke(std::forward<F>(f), std::move(value()));
}
else {
    return std::move(*this);
}
```

1 `template <class F> constexpr auto transform(F&& f) &;`

2 `template <class F> constexpr auto transform(F&& f) const&;`

Let `U` be `invoke_result_t<F, decltype(value())>`.

Constraints: F models `invocable<decltype(value())>`. E and U model `copy_constructible`.

Effects: Equivalent to:

```
if (*this) {  
    return expected<U,E>(in_place, invoke(std::forward<F>(f), value()));  
}  
else {  
    return *this;  
}
```

1 `template <class F> constexpr auto transform(F&& f) &&;`

2 `template <class F> constexpr auto transform(F&& f) const&&;`

Let U be `invoke_result_t<F, decltype(std::move(value()))>`.

Constraints: F models `invocable<decltype(std::move(value()))>`. E and U model `move_constructible`.

Effects: Equivalent to:

```
if (*this) {  
    return expected<U,E>(in_place, invoke(std::forward<F>(f), std::move(  
}  
else {  
    return std::move(*this)  
}
```

1 `template <class F> constexpr expected_or_else(F&& f) const&;`

Constraints: F models `invocable<>` and E models `copy_constructible`.

Mandates: `is_same_v<remove_cvref_t<invoke_result_t<F>>, expected>` is true.

Effects: Equivalent to:

```
if (*this) {  
    return *this;  
}  
else {  
    return invoke(std::forward<F>(f)(error()));  
}
```

```
1 template <class F> constexpr expected_or_else(F&& f) &&;
```

Constraints: F models `invocable<>` and E models `move_constructable`

Mandates: `is_same_v<remove_cvref_t<invoke_result_t<F>>, expected>` is true.

Effects: Equivalent to:

```
if (*this) {  
    return std::move(*this);  
}  
else {  
    return invoke(std::forward<F>(f)(std::move(error())));  
}
```

Partial specialization of `expected` for void types `[expected.void]`

After .8.6, observers – add

```
// 0.0.8.7, monadic
template <class F> constexpr auto and_then(F&& f) &;
template <class F> constexpr auto and_then(F&& f) &&;
template <class F> constexpr auto and_then(F&& f) const&;
template <class F> constexpr auto and_then(F&& f) const&&;
template <class F> constexpr expected_or_else(F&& f) &&;
template <class F> constexpr expected_or_else(F&& f) const&;
```

Partial specialization of expected for void [expected.void.observe]

After section 8.5 Observers [expected.void.observe]

Add section 8.6 Monadic [expected.void.monadic]

```
1 template <class F> constexpr auto and_then(F&& f) &;
```

```
2 template <class F> constexpr auto and_then(F&& f) const&;
```

Constraints: F models `invocable<>` and E models `copy_constructible`.

Mandates: `is_same_v<remove_cvref_t<invoke_result_t<F>>, expected>` is true.

Effects: Equivalent to:

```
if (*this) {
    return invoke(std::forward<F>(f)());
}
else {
    return *this;
}
```

```
1 template <class F> constexpr auto and_then(F&& f) &&;
```

```
2 template <class F> constexpr auto and_then(F&& f) const&&;
```

Constraints: F models `invocable<>` and E models `move_constructible`.

Mandates: `is_same_v<remove_cvref_t<invoke_result_t<F>>, expected>` is true.

Effects: Equivalent to:

```
if (*this) {  
    return invoke(std::forward<F>(f)());  
}  
else {  
    return std::move(*this);  
}
```

```
1 template <class F> constexpr expected_or_else(F&& f) const&;
```

Constraints: F models `invocable<>` and E models `copy_constructible`.

Effects: Equivalent to:

```
if (*this) {  
    return *this;  
}  
else {  
    return invoke(std::forward<F>(f)());  
}
```

```
1 template <class F> constexpr expected_or_else(F&& f) &&;
```

Constraints: F models `invocable<>` and E models `move_constructible`.

Effects: Equivalent to:

```
if (*this) {  
    return std::move(*this);  
}  
else {  
    return invoke(std::forward<F>(f)());  
}
```

Acknowledgements

- Thanks to Sy Brand for the original work on optional and expected.
- Thanks to Broneck Kozicki for early feedback.

Revision History

Version	Date	Changes
0	2021-12-12	Initial Draft

References

1. [P0798] Sy Brand "Monadic Functions for std::optional"
<https://wg21.link/P0798>
2. C++ draft [optional.nomadic] <http://eel.is/c++draft/optional.monadic>
3. [P0323] Vicente Botet, JF Bastien, Jonathan Wakely std::expected
<https://wg21.link/P0323>
4. Sy Brand expected <https://github.com/TartanLlama/expected>
5. Sy Brand expected docs
<https://tl.tartanllama.xyz/en/latest/api/expected.html#tl-expected>
6. Rust Result <https://doc.rust-lang.org/std/result/enum.Result.html>
7. More complete examples <https://godbolt.org/z/va5fzx11f>